

C1 · Parsing-Retrieval Disconnect

Candidate parsers



Qwen3-VL



MinerU



Marker



PaddleOCR

Cleaner-looking ≠ better retrieval

parser	Boundary Clarity	Hit@1
MinerU	0.72 (highest)	0.20
Prod (ours)	0.61	0.55

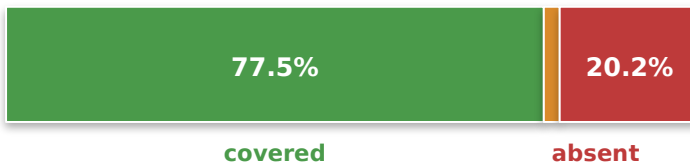
parser choice alone moves Hit@1 by **+35.1pp**
(0.20→0.55; 2.8×)

appearance metrics measure formatting, not content
(anti-correlation $r = -0.81$, $n = 5$)

C2 / C3 · Diagnose + Select

C2 · Coverage diagnostic (no retriever)

where does each gold answer land?



absent = answer never made it into the parser output
(parser-side failure)
set before chunking → no chunker can recover it

C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3



mE5



Qwen3-Emb

→ rank parsers & chunkers by retrieval

C4 · Bounded Parser Training

C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

RADP-Distill — edit-distance → ref

RADP-DPO — page-local RCPS

Result

Hit@5 gain over the untuned parser:

RADP-Distill  **+1.22 pp**

RADP-DPO  **+0.85 pp**

substantially overlapping CIs
no evidence reward helps; lever = fidelity distillation